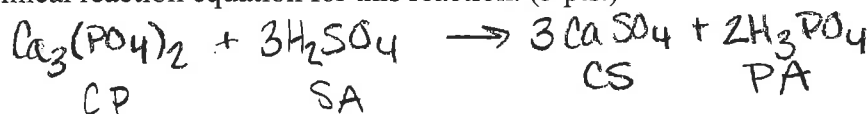


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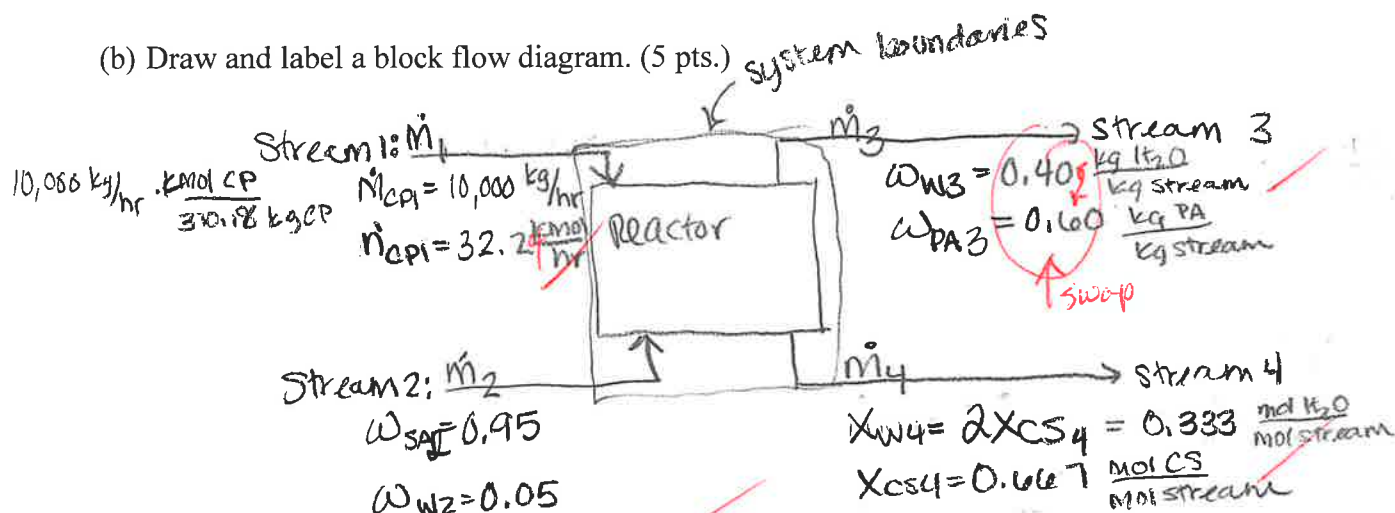
- (1) (50 pts.) Phosphoric acid (PA) can be produced by reacting calcium phosphate (CP) with a stoichiometric quantity of sulfuric acid (SA). Calcium sulfate (CS) is also a byproduct of this reaction. Two streams enter the reactor: (**Stream 1**) has a species mass flow rate of 10,000 kgCP/hr of CP which is suspended in an unknown amount of water (W) and (**Stream 2**) a 95 wt% aqueous solution of SA. There are two exit streams: (**Stream 3**) a 40 wt% aqueous solution of PA and (**Stream 4**) a solid composed of CS and water which has a mole ratio of 2 moles of water for each mole of CS.

Species	Molar Mass
CP ($\text{Ca}_3(\text{PO}_4)_2$)	310.18
SA (H_2SO_4)	98.08
PA (H_3PO_4)	97.994
CS (CaSO_4)	136.14
W (H_2O)	18.02

- (a) Write a balanced chemical reaction equation for this reaction. (5 pts.)



- (b) Draw and label a block flow diagram. (5 pts.)



- (c) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. In the table below, give values for 4a, 5a, 5b, 5c, and 5d along with a brief explanation for the value you give. Notice that the value for 4b is given with an explanation. Compute the DOF. (10 pts.)

	Number	Explanation
4a. Stream Variables	8	Each stream contains 2 species $\times 4 = 8$
4b. System Variable	0	No reactants in product streams. The reaction is complete.
Total Independent Variables	89	
5a. Specified Flows	1	$\dot{m}_{\text{CP1}}$ species flow rate
5b. Specified Stream Compositions	3	(N-1) for streams 2, 3, 4
5c. Specified System Performances	0	No reactants in product streams or accumulated in the reactor
5d. Material Balance Equations	5	CP, SA, PA, CS, W
Total Independent Equations	9	
DOF	80	overspecified

[Problem 2 continued]

- (c) (30 pts.) Calculate the species molar flow rates of the ammonia leaving the system, that is, A_2 and A_4 . This calculation is likely to require solving a quadratic equation.

Species	Stream 1		Stream 2		Stream 3		Stream 4	
	x_n	$\dot{n}$	x_n	$\dot{n}$	x_n	$\dot{n}$	x_n	$\dot{n}$
N_2	0.9	33.45	—	—	$1-2x_{4A}$	33.45	—	—
NH_3	0.1	3.72	—	—	$2x_{4A}$	2.038	x_{4A}	—
H_2O	—	—	1	55.6	—	—	$1-x_{4A}$	55.6
Total	1.0	37.17	1	55.6	1	35.49	1	57.24
	input				output			

$$\dot{n}_2 = \frac{\dot{n}_{2N}}{x_{2N}} = \frac{33.45}{1-2(0.0287)}$$

$$= 35.49 \frac{\text{mol } a}{\text{hr}}$$

$$\dot{n}_4 = \frac{\dot{n}_{4W}}{1-x_{4A}} = \frac{55.6}{1-0.0287} = 57.24$$

Ammonia

$$3.72 = 2x_{4A} \dot{n}_2 + x_{4A} \dot{n}_4$$

Nitrogen

$$33.45 = (1-2x_{4A}) \dot{n}_2$$

Water

$$55.6 = (1-x_{4A}) \dot{n}_4$$

$$3.72 = 2x_{4A} \left(\frac{33.45}{1-2x_{4A}} \right) + x_{4A} \left(\frac{55.6}{1-x_{4A}} \right)$$

$$3.72 = \frac{66.9x_{4A}}{1-2x_{4A}} + \frac{55.6x_{4A}}{1-x_{4A}}$$

$$3.72(1-2x_{4A})(1-x_{4A}) = 122.5x_{4A}$$

$$1 - 3x_{4A} + 2x_{4A}^2 = 32.96x_{4A}$$

$$2x_{4A}^2 - 35.96x_{4A} + 1 = 0$$

$$b^2 - 4AC = (-35.96)^2 - 4(2)(1) = 1282.96$$

$$\sqrt{b^2 - 4AC} = 35.84$$

Check

$$\dot{n}_{1A} = \dot{n}_{2A} + \dot{n}_{4A} = 2.038 + 1.643$$

$$3.72 \neq 3.681$$

$$N_2 = 33.45 \text{ kmolN/hr } A_2 = 2.038 \text{ kmolA/hr } W_4 = 55.6 \text{ kmolW/hr } A_4 = 1.643 \text{ kmolA/hr}$$

$$x_{4A} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{35.96 \pm 35.84}{2(2)}$$

$$= 17.94 > 1$$

or

$$= 0.0287$$

x_{4A} is a mole fraction $\hat{x}_i$ should be less than 1

If we round $3.7 = 3.7$

$$\dot{n}_{A2} = x_{A2} \dot{n}_2 = 2(0.0287)(35.49 \frac{\text{mol}}{\text{hr}})$$

$$= 2.038 \frac{\text{mol } NH_3}{\text{hr}}$$

$$\dot{n}_{A3} = x_{A3} \dot{n}_3 = (0.0287)(57.24) = 1.643 \frac{\text{molA}}{\text{hr}}$$

- (d) (5 pts.) What is the percent recovery of the ammonia, that is, what percentage of the ammonia entering the separator is removed with the water stream?

$$\left(\frac{2.038 \text{ mol } A_2}{3.72 \text{ mol } A_1} \right) \times 100 = 54.8\%$$

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NAME Gillianne Murdock

- (2) (50 pts.) A gas mixture of nitrogen N_2 (N) and ammonia NH_3 (A) enters the bottom of a separator as **Stream 1** with mass flow rate $\dot{m}_1 = 1000$ kg/hr and a mole fraction of the ammonia $x_{A1} = 0.10$. **Stream 2** exits the top of the separator containing a gas mixture of nitrogen and ammonia. Pure liquid water H_2O (W) enters the top of the same separator as **Stream 3** at a mass flow rate of $\dot{m}_3 = 1000$ kg/hr and a liquid mixture of water and ammonia exits the bottom as **Stream 4**. The nitrogen and water do not mix during the process. The mole fractions of the ammonia in the two exiting streams are known to be related by $x_{2A} = 2x_{4A}$. Assume steady state. Molecular Weights: $M_A = 17$ kg/kmol for ammonia, $M_N = 28$ kg/kmol for nitrogen, and $M_W = 18$ kg/kmol for water.

No accumulation

- (a) (5 pts.) Calculate the mass fraction of A in Stream 1, that is w_{1A} .

$$x_{N1} = 0.9$$

$$x_{A1} = 0.1$$

$$0.9 \text{ mol } N_2 = 25.2 \text{ g } N_2$$

$$0.1 \text{ mol } NH_3 = 1.7 \text{ g } NH_3$$

$$1 \text{ mol mix} = 26.9 \text{ g mixture}$$

$$w_{A1} = \left(\frac{1.7}{26.9} \right) \times 100 = 6.32\%$$

$$w_{1A} = 0.0632 \text{ kgA/kg}$$

- (b) (10 pts.) Calculate the species molar flow rates for the inlet ammonia A_1 , nitrogen N_1 , and water W_3 .

$$\dot{n}_1 = 1000 \frac{\text{kg}}{\text{hr}} \cdot \frac{1 \text{ kmol mix}}{26.9 \text{ kg mix}} = 37.17 \frac{\text{kmol mix}}{\text{hr}}$$

$$\dot{n}_{A1} = \left(0.1 \frac{\text{kmol A}}{\text{kmol mix}} \right) \left(37.17 \frac{\text{kmol mix}}{\text{hr}} \right) = 3.72 \frac{\text{kmol A}}{\text{hr}}$$

$$\dot{n}_{N1} = 37.17 \frac{\text{kmol mix}}{\text{hr}} - 3.72 \frac{\text{kmol A}}{\text{hr}} = 33.45 \frac{\text{kmol N}}{\text{hr}}$$

$$\dot{m}_3 = \dot{m}_4$$

$$\dot{n}_3 = \frac{\dot{m}_3}{M_W} = \frac{1000 \frac{\text{kg } H_2O}{\text{hr}}}{18 \frac{\text{kg } H_2O}{\text{kmol}}} = 55.6 \frac{\text{kmol } H_2O}{\text{hr}}$$

$$\dot{n}_{A1} = \dot{n}_{A2} + \dot{n}_{A4}$$

$$\begin{aligned} \dot{n}_{A1} &= x_{A2} \dot{n}_2 + x_{A4} \dot{n}_4 \\ &= 2x_{A4} \dot{n}_2 + x_{A4} \dot{n}_4 \\ &= x_{A4} (2\dot{n}_2 + \dot{n}_4) \end{aligned}$$

$$\dot{n}_{N1} = \dot{n}_{N2} = 33.45 \frac{\text{kmol N}}{\text{hr}}$$

$$\begin{aligned} \dot{m}_{N2} &= \dot{n}_{N2} \cdot M_N \\ &= \left(33.45 \frac{\text{kmol}}{\text{hr}} \right) \left(28 \frac{\text{kg N}}{\text{kmol}} \right) \\ &= 936.6 \frac{\text{kg N}}{\text{hr}} \end{aligned}$$

$$\dot{m}_2 =$$

$$\frac{1000 \text{ kg mix}}{\text{hr}} \cdot 0.0632 \frac{\text{kg A}}{\text{kg mix}} = 63.2 \frac{\text{kg A}}{\text{hr}} \cdot \frac{1 \text{ kmol A}}{17 \text{ kg A}}$$

$$= 3.72 \frac{\text{kmol A}}{\text{hr}}$$

$$A_1 = 3.72 \text{ kmolA/hr} \quad N_1 = 33.45 \text{ kmolN/hr} \quad W_3 = 55.6 \text{ kmolW/hr}$$

[Problem 1 continued]

- (d) Calculate the species molar flow rates for the four non-water components in the four streams in kmol/hr. That is CP_1 , SA_2 , PA_3 , and CS_4 where the subscripts are stream numbers. Also, write a balance on water and calculate W_1 , the water entering the process with CP_1 . Show all your calculations and write your final answers in the spaces provided below. (30 pts.)

$$CP_1 = \frac{\dot{m}_{CP1}}{M_{CP}} = \frac{10,000 \text{ kg CP/hr}}{310.18 \text{ kg CP/kmol}} = 32.2 \text{ kmol/hr}$$

$$CS_4 = \frac{32.2 \text{ kmol CP}}{\text{hr}} \left| \frac{3 \text{ kmol CS}}{1 \text{ kmol CP}} \right| = 96.6 \text{ kmol CS/hr}$$

$$PA_3 = \frac{32.2 \text{ kmol CP}}{\text{hr}} \left| \frac{2 \text{ kmol PA}}{1 \text{ kmol CP}} \right| = 64.4 \text{ kmol PA/hr}$$

$$SA_2 = \frac{32.2 \text{ kmol CP}}{\text{hr}} \left| \frac{3 \text{ kmol SA}}{1 \text{ kmol CP}} \right| = 96.6 \text{ kmol SA/hr}$$

$$\begin{array}{r} 1 \ 48 \\ 2 \ 47 \\ \hline 1 \ 95 \end{array}$$

Water:

$$\begin{aligned} \dot{m}_{PA3} &= (\dot{n}_{PA3})(M_{PA}) \\ &= (64.4 \text{ kmol PA/hr}) \left(\frac{97.994 \text{ kg PA}}{\text{kmol PA}} \right) \\ &= 6310.81 \text{ kg PA/hr} \end{aligned}$$

$$\dot{m}_3 = \frac{\dot{m}_{PA3}}{PA_3} = \frac{6310.81}{0.604} = 10,518 \text{ kg/hr}$$

$$\dot{m}_{3W} = (0.4)(10,518 \text{ kg/hr}) = 4207.21 \text{ kg/hr}$$

$$\dot{n}_{3W} = \frac{4207.21 \text{ kg H}_2\text{O}}{\text{hr}} \cdot \frac{1 \text{ kmol H}_2\text{O}}{18.02 \text{ kg H}_2\text{O}} = 233.47 \text{ kmol H}_2\text{O/hr}$$

$$\dot{m}_{SA2} = (\dot{n}_{SA2})(M_{SA}) = (96.6 \text{ kmol SA/hr}) \left(\frac{48.08 \text{ kg SA}}{\text{kmol SA}} \right) = 4644.53$$

$$\dot{m}_2 = \frac{\dot{m}_{SA2}}{WSA2} = \frac{4644.53}{0.95} = 4888.98 \text{ kg/hr}$$

$$\dot{m}_{2W} = (\dot{m}_2)(\omega_{W2}) = (4888.98 \text{ kg/hr})(0.05) = 244.45 \text{ kg H}_2\text{O/hr}$$

$$\dot{n}_{2W} = \frac{\dot{m}_{2W}}{M_W} = \frac{244.45 \text{ kg H}_2\text{O}}{\text{hr}} \cdot \frac{1 \text{ kmol H}_2\text{O}}{18.02 \text{ kg H}_2\text{O}} = 13.56 \text{ kmol H}_2\text{O/hr}$$

$$\dot{n}_4 = \frac{\dot{n}_{CS4}}{X_{CS4}} = \frac{96.6 \text{ kmol CS/hr}}{0.663} = 144.90 \text{ kmol Str/hr}$$

$$\begin{aligned} \dot{n}_{4W} &= X_{4W} \cdot \dot{n}_4 \\ &= (0.333)(144.90 \text{ kmol/hr}) = 48.30 \text{ kmol H}_2\text{O/hr} \end{aligned}$$

$$\dot{n}_{1W} + \dot{n}_{2W} = \dot{n}_{3W} + \dot{n}_{4W}$$

$$\begin{aligned} \dot{n}_{1W} &= \dot{n}_{3W} + \dot{n}_{4W} - \dot{n}_{2W} \\ &= (233.47 + 48.30 - 13.56) \text{ kmol/hr} \\ &= 268.21 \text{ kmol H}_2\text{O/hr} \end{aligned}$$

$$\begin{array}{c} 28 \\ \text{48} \end{array}$$

Species molar flow rate. Subscripts indicate stream numbers.

CP_1 (kmol/hr)	SA_2 (kmol/hr)	PA_3 (kmol/hr)	CS_4 (kmol/hr)	W_1 (kmol/hr)
32.2	96.6	64.4	96.6	268.21